The Bessel-Moment Period Algebra of the Three-Center Coulomb Integral: Modular Structure and Cosmic-Galois Placement*

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(Dated: August 16, 2026)

The companion Paper 59 [22] establishes that the genuine three-center electron-repulsion integral over $1s$ Slater orbitals, read in momentum space under Fock's conformal projection, is a two-scale elliptic Bessel moment: its radial content is a period of the rank-four irregular Picard–Fuchs connection of the genus-one curve $y^2 = (c_1k^2+1)(c_2k^2+1)$, and its finite closed form is open. This paper takes that object as its starting point and develops its number-theoretic structure. **[MEASURED; classical]** The elliptic family is the Legendre family over the modular curve $X(2) = \Gamma(2)\backslash\mathfrak{H}^*$, pulled back along the exact linear map $\lambda(\tau(\rho)) = 1-\rho$; the parameter integral sweeps the physical arc $\lambda < 1$ of the real locus of $X(2)$ and passes through complex-multiplication fibres—at $\tau = i$ ($\rho = \frac{1}{2}$, inside the physical domain) and $\tau = i\sqrt{2}$—where the periods are Chowla–Selberg Γ-values, so a molecular integral carries classical Γ-value periods across distinct fundamental discriminants. **[OBSERVATION]** The integrated observable lives on $X_0(2)$, is a $\Gamma(2)$ resurgent Lambert series (the pure multiple modular value being its regular $D \to 0$ shadow), and carries conductor-4 ($\mathbb{Z}[i]$, $G = \beta(2)$) arithmetic through the fibre theta series $K = \frac{\pi}{2}\theta_3^2$. That modular and complex-multiplication structure—not the integer-relation search—places it inside the cosmic-Galois / mixed-elliptic-motive layer the program's Tannakian substrate reaches for; the guarded, decoy-calibrated searches are separately and decisively negative for a clean low-height closed form through weight three. **[SYMBOLIC + MEASURED]** The connection has four master periods whose Wronskian is the pure monomial W_0D^{-2} and whose Lagrange bilinear concomitant—the analogue, here for the $\Gamma(2)$ (Legendre) family, of the Broadhurst–Roberts / Fresán–Sabbah–Yu quadratic period relations, whose proved case is the Kloosterman connection—is identified in closed form as $B = \pi\,\Omega$, with Ω the canonical integer symplectic intersection form, so the *monodromy* group lies in $\mathrm{Sp}_4(\mathbb{Z})$ (the differential Galois group, being a positive-dimensional algebraic group here, lies in $\mathrm{Sp}_4(\mathbb{C})$). The object is an Eisenstein / complex-multiplication Bessel-moment period, neither a cusp-form L-value nor an elliptic dilogarithm; the explicit finite reduction remains the open frontier.

I. INTRODUCTION: THE OBJECT, AND THE TOWER ABOVE IT

The genuine three-center electron-repulsion integral $(XY|XZ)$ over $1s$ Slater orbitals is, read in momentum space under Fock's conformal projection, a two-scale elliptic Bessel moment: a period of the rank-four irregular Picard–Fuchs connection of the genus-one curve $y^2 = (c_1k^2+1)(c_2k^2+1)$, whose two scales are the Feynman/Källén–Lehmann parameters $c_1 = s(1-s)$ and $c_2 = t(1-t)$. The construction of that object—the momentum reduction, the transcendence jump from the genus-zero two-center class, the elliptic-dilogarithm obstruction, and the differential-Galois irreducibility of the connection—together with its chemistry context and full numerical validation is the companion Paper 59 [22], which this paper takes as established and as its starting point.

Here we develop the number-theoretic tower that sits above it. Section II places the elliptic family as the Legendre family over $X(2)$, exhibits its rational modulus map and its in-domain complex-multiplication fibres, and reads the integrated observable as a $\Gamma(2)$ resurgent Lambert series carrying conductor-4 arithmetic—its cosmic-Galois placement. Section III makes that placement quantitative through the period algebra of the master family: the Wronskian W_0D^{-2}, the quadratic period relations, and the intersection form $B = \pi\,\Omega$ that forces the *monodromy* group into $\mathrm{Sp}_4(\mathbb{Z})$, and the differential Galois group into $\mathrm{Sp}_4(\mathbb{C})$.

Throughout, "Paper 59" denotes the companion; the equations and sections of Paper 59 that this development rests on (the genus-one curve, the rank-four Picard–Fuchs connection and its one-mass slice $N(D)$, the two-mass Bessel moment, and the obstruction analysis) are pointed to in prose rather than by cross-reference. The tier tags (**[SYMBOLIC]**, **[MEASURED]**, **[OBSERVATION]**, **[OPEN]**) are carried over from the companion unchanged.

Transcendence bookkeeping. **[OBSERVATION]** The transcendentals below are placed against the program's taxonomy (Papers 18, 34 [18, 19]), as the companion Paper 59 places its own. Paper 18's *embedding* tier carries a *genus* grading, and this object sits one rung above the *two-centre* class: the two-center constants are the genus-zero set $\{E_1, \ln, \gamma_E\}$, while the three-center fibre — which this paper shares with its companion, whose own title places it at genus one — leaves that tower for the elliptic one. Concretely: π enters at the measure tier of Paper 34 and appears here as the overall factor of the

* Paper 61 in the Geometric Vacuum series

intersection form $B = \pi\Omega$ and in $K = \frac{\pi}{2}\theta_3^2$; the Jacobi theta constants $\theta_2, \theta_3, \theta_4$ and the constant $\varpi \equiv K(\frac{1}{2})$ (this corpus's symbol; the classical lemniscate constant is $\sqrt{2}\,\varpi$) are *fibre* data of the Legendre family, not projection constants; the Γ-values $\Gamma(1/4)^2$ and $\Gamma(1/8)\Gamma(3/8)$ are Chowla–Selberg periods attached to the complex-multiplication fibres and are therefore *arithmetic* rather than analytic in origin; ζ and β appear only through the factorization $L(\theta_3^2, s) = 4\zeta(s)\beta(s)$ of the $\mathbb{Z}[i]$ theta series, so Catalan's constant $G = \beta(2)$ is conductor-4 *family* data; and γ_E appears only as coordinate bookkeeping in the weight-one seed set (§II), a reading this paper records as an observation rather than a re-tiering of Paper 18's ownership of that constant. The load-bearing consequence is the one the searches measure: *no* measured conductor-4 content ($\beta(2)$) survives in the integrated physical value at reachable heights, so the arithmetic lives in the family and in the $D \to 0$ shadow rather than in the observable.

II. THE MODULAR STRUCTURE AND ITS COSMIC-GALOIS PERIODS

The genus-one curve of Paper 59 is not an arbitrary elliptic curve: the family over the scale-ratio base is a *modular* family, and its periods are periods of a specific modular curve. Forming the modular parameter from the two periods of Paper 59, $\tau(\rho) = iK(\rho)/K(1-\rho)$, the theta-series modular lambda satisfies **[MEASURED, precision-limited (**2.7×10^{-51} **at 50 digits,** 1.2×10^{-91} **at 90); classical]**

$$\lambda\bigl(\tau(\rho)\bigr) = 1 - \rho, \tag{1}$$

an exact, rational—indeed linear—modulus map. The family is therefore the Legendre family, the universal elliptic curve over the modular curve $X(2) = \Gamma(2)\backslash\mathfrak{H}^*$, pulled back along $\lambda = 1 - \rho$; it is non-isotrivial ($j(\rho)$ varies) with three cusps at $\rho \in \{0, 1, \infty\}$ matching the three cusps of $\Gamma(2)$ (the cusp $\rho = 1$ is the coincident-scale diagonal). The periods $K(1-\rho), K(\rho)$ are the $\Gamma(2)$ modular periods.

The modulus is a four-point cross-ratio. Identity (1) has an elementary source. Under $u = x^2$ the quartic model reduces $2:1$ to the elliptic curve $v^2 = u(u-1)(\rho u + 1 - \rho)$, whose periods are $K(\rho), K(1-\rho)$ and whose four branch points are $\{0, 1, \infty, (\rho-1)/\rho\}$. Their six-element cross-ratio orbit is $\{(\rho-1)/\rho,\ 1/\rho,\ \rho/(\rho-1),\ \rho,\ 1-\rho,\ -1/(\rho-1)\}$ **[SYMBOLIC]**, and it contains *both* period arguments ρ and $1-\rho$; the Legendre value $1-\rho$ is one of its members. The elliptic modulus of the three-center integral is therefore the classical modulus of four marked points on $\mathbb{P}^1$—the tame four-subspace ($\tilde{D}_4$) modulus, equivalently the cross-ratio of the four regular singular points of a rank-two Fuchsian system (the isomonodromic setting whose deformation is Painlevé VI). One caveat fixes the reading: the four points are the branch data of the *two* spectral masses c_1, c_2 through $\rho = c_2/c_1$ (each mass contributes a $\pm$ pair, one pair sent to $\{0, \infty\}$ and the other to $\{1, (\rho-1)/\rho\}$ by the reduction), *not* the four positions of four scattering centers—the modulus counts the two densities, not the number of nuclei.

Since $\rho = c_2/c_1$ ranges over all of $(0, \infty)$ as the Feynman parameters range over $(0, 1)$, the parameter integral sweeps the arc $\lambda < 1$ of the real locus of $X(2)$ — the physical range $\rho \in (0, \infty)$ gives $\lambda = 1 - \rho < 1$, so the arc $\lambda > 1$ is not reached — and passes through infinitely many of its complex-multiplication fibres (not every one: those with non-real λ, such as discriminant -3, are off the contour, as is $\lambda = 2$; see below in this section). There the periods are Γ-values by Chowla–Selberg **[MEASURED,** $\sim 10^{-51}$**; classical]**: at $\tau = i$ (the fibre $\rho = \frac{1}{2}$, *inside* the physical domain), $K(\frac{1}{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$; at $\tau = i\sqrt{2}$ (discriminant -8), $K = (1+\sqrt{2})^{1/2}\Gamma(1/8)\Gamma(3/8)/(2^{13/4}\sqrt{\pi})$. A molecular three-center integral thus has, at the fibres its own parameter integral visits, periods that are classical Γ-values—across distinct fundamental discriminants, so the arithmetic tracks the fibre rather than a single coincidence.

The integrated integral is an elliptic Bessel moment. The full parameter integral pulls back, via Eq. (1), to an iterated integral over $X(2)$ of period-weighted forms. The fibre, however, carries an irregular exponential (Bessel) twist—it degenerates to the Bessel value $K_0(1)$ as $\rho \to 0$, which is not a modular period—so the object is properly a *Bessel moment of the Legendre family*, an irregular/exponential period in the sense of Fresán–Sabbah–Yu [8], of which the $\Gamma(2)$ multiple modular value in the sense of Brown [9] (an elliptic polylogarithm [11, 12]) is the regular $D \to 0$ shadow. A dedicated evaluator—a fixed Gauss–Legendre tensor grid with a $\sin^2$ substitution on the smooth bulk and, at the $(s, t) \to (0, 0)$ corner, a Duffy angular split followed by the radial substitution $\rho = \sigma^2$ ($\rho = s + t$) that renders the $\rho^{3/2}$ corner integrand *analytic* in σ and so restores spectral Gauss–Legendre convergence—gives the collinear value **[MEASURED, 18 digits confirmed]** $0.3953557659017139641\ldots$ (correct to 18 digits; the 19th is superseded by the certified value below). The earlier $\sim$16-digit figure was not an intrinsic wall but an under-resolution of the corner and of the one wide sub-domain flagged above (both closed here; the corner by $\rho = \sigma^2$, the sub-domain by cross-validated Gauss–Legendre and tanh–sinh quadrature); the $\sim$20-digit ceiling was a fixed-fibre-grid artifact, not intrinsic: on the decay-scaled map $k = Lu/(1-u)$, $L = 1/(\sqrt{c_1} + \sqrt{c_2})$, a fixed Gauss–Legendre fibre is *spectral* to 40-plus digits (the fibre is not the bottleneck), and a rebuilt spectral evaluator ($\rho = \sigma^2$ radial + $\sin^2$ angular corner, kink-free trapezoid split) independently reproduces the value to 16 digits (the two structurally distinct evaluators agree to 16 digits; a naive single-axis outer-grid push produces non-monotonic near-coincidences past that and is not trusted); the residual precision limit is the outer-integral

Gauss–Legendre rate ($\sim e^{-0.55N}$, complex-singularity-limited), so within the (s,t) frame a definitive evaluation required a different representation—supplied below by the exact (k,w) refactorization, Eq. (2), not by the modular route. [**MEASURED**] A guarded integer-relation search with a same-magnitude decoy control finds the value is *not* a low-height combination of the small disc-4 complex-multiplication ring $\{1, \pi, K(\frac{1}{2}) = \Gamma(\frac{1}{4})^2/4\sqrt{\pi}\}$ at weight ≤ 2: the relation height is far above low and is matched by the decoy, the signature of no low-height relation—and this leg is decisive, since a weight-≤ 2 disc-4 basis is small enough to resolve at the reachable precision. The wider searches, through weight three and against the disc-8-inclusive ring $\{\pi, K(\frac{1}{2}), P_8\}$, are only *consistent* with the negative at present: those bases are over-determined at ~ 19 digits—the spurious low-height hits differ from one precision to the next, and are matched by the decoy—a limitation removed below. All of this is consistent with a genuine (non-classical, multi-fibre) elliptic Bessel-moment period.

[**SYMBOLIC + MEASURED**] Both the precision wall and the pending weight-three decision are closed by an exact refactorization. Writing $j_0(kb) = \int_0^1 \cos(kbw)\,dw$ decouples the $b = s+t$ phase, and the reflection symmetry $P(s,k) = P(1-s,k)$ of the collinear packet makes the inner integral real, so the (s,t) double integral collapses to the *square of a one-dimensional integral*:

$$T2 = \frac{8}{\pi}\int_0^\infty dk \int_0^1 dw\, \cos(kw)\, R(k,w)^2, \quad R(k,w) = \int_0^1 \cos\left(kw(s-\tfrac{1}{2})\right) P(s,k)\, ds. \tag{2}$$

In this frame the complex off-axis (s,t) singularity that produced the $\sim$13–14-digit outer wall does not exist (the k-integrand is entire in w; the s-integrand is analytic on $[0,1]$), no fibre is ever evaluated, and the entire precision question is the closed-form large-k tail (a Watson expansion; the Gevrey structure relocates to a truncation floor $e^{-K/2}$ with the cut K freely chosen). Six parameter-disjoint runs—two independent parallel configurations agreeing to 1.7×10^{-67}, the identity itself verified against a fully independent two-dimensional evaluation to 69–96 digits—certify

$$T2 = 0.395355765901713964325229296804847564260563977867082108935234265469\ldots$$

(66 digits; a *decomposed* certification—identity, quadrature and tail legs each closed independently, at higher precision than any single end-to-end pipeline; `tests/test_paper59_t2_value.py` pins the identity, the K^{-7} tail law, and the digit-19 correction of the frozen anchor above, and the certified entry with its full evidence accounting is in the reference artifact, `benchmarks/certified_reference/`). At this precision the guarded, decoy-calibrated integer-relation searches become decisive across the board: the corrected weight-≤ 3 ring $\{\pi, K(\frac{1}{2})^{\pm1}, G\}$ is **decisively negative** at height ≤ 10 (64 working digits, two precisions), as are disc-4 through weight three and disc-8 through weight two at heights up to 10^4–10^{12}, and eight targeted Catalan probes—*no measured conductor-4 ($\beta(2)$/Catalan) content exists in the integrated observable at clean heights.* This sharpens the shadow reading (the finite weight-three-requiring-G element belongs to the $D\to0$ shadow, not to $T2$) and answers the Paper 56 seam test T-2 in the negative at the clean-height bound. Two calibration riders: a decision on the dimension-twenty ring was never available at the previously projected $\sim$32–40 digits (height budgets scale as $10^{D/n}$ in the ring dimension n), and the negative excludes *clean* closed forms, not large-height ones. The resurgent trans-series program below is untouched—digits and closed form are separate questions; the closed form remains [OPEN].

[**MEASURED**] Two refinements sharpen this. *First*, the fibre precision wall is removable. The edge oscillation of $j_0(k(s+t))$—which runs over a k-range $\sim 1/\sqrt{c_{\min}}$ as a Fock scale collapses and limited every fixed- and decay-grid fibre quadrature to $\sim$16–22 digits—is eliminated by an *oscillation-free* evaluator: split the radial integral as $\int_0^K + \int_K^\infty$ and evaluate the oscillatory tail $\int_K^\infty \sin(kb)\, e^{-Ak}\, k^{-n}\, dk = \mathrm{Im}\big[z^{\,n-1}\Gamma(1-n, zK)\big]$ in closed form ($z = A - ib$, $A = \sqrt{c_1} + \sqrt{c_2}$, $b = s+t$), the tail amplitudes taken from the $1/k$-series of the cancellation-stabilised combination $g(k) = P_1(k)P_2(k)\, e^{Ak} k^6$ (a single $\Gamma(-6, zK)$ plus downward recurrence). This reproduces the decay-grid fibre to 40–45 digits at every (s,t), the corner included, and reproduces the collinear value to its confirmed digits. With the fibre no longer the bottleneck, the residual precision wall of that frame was the *outer* (s,t)-integral's corner-limited convergence—which held the value near a $\sim$19-digit ceiling there until the (k,w) refactorization of Eq. (2) dissolved the question—not the fibre. This outer wall is now characterized: of the four corners only $(s,t)\to(0,0)$ is fractionally singular ($\rho^{3/2}$, absorbed by the $\rho=\sigma^2$ Duffy)—the three oscillatory corners ($b = s+t \neq 0$) are integer-leading (leading order ρ^2) and carry no fractional-power subtraction [**MEASURED;** `test_paper59_resurgence_corner.py`]—so the cap is a weak *complex off-axis* singularity of J (its signature a non-monotone Gauss–Legendre rate), not a missing corner map; a third, coverage-corrected momentum-space evaluator independently reproduces the value to $\sim$13–14 digits and then plateaus, confirming the anchor's leading digits; no (s,t)-frame quadrature reached further, and the definitive digits were ultimately supplied by the (k,w) refactorization (Eq. (2)), not by the modular route. *Second*, the guarded searches above used a *period-only* ring. A length-two, weight-≤ 3 iterated integral of weight-two Eisenstein series over $X(2)$—the shape forced by the two Feynman integrations—generically produces also the Legendre *quasiperiod* $1/K(\frac{1}{2})$ (the second-kind companion, $E(\frac{1}{2}) = \pi/4K(\frac{1}{2}) + K(\frac{1}{2})/2$) and the weight-two Eisenstein L-value $G = \beta(2) = L(2,\chi_{-4})$ (Catalan's constant; native to this family because the fibre period *is* the $\mathbb{Q}(i)$ theta series: $K(\lambda) = \frac{\pi}{2}\theta_3^2$

with $\theta_3^2 = 1 + \sum_{n\geq 1} r_2(n)\, q^n$, $r_2(n) = 4\sum_{d|n}\chi_{-4}(d)$ by Jacobi's two-squares theorem—the weight-one, level-4 Eisenstein series of character χ_{-4}, with $L(\theta_3^2, s) = 4\,\zeta(s)\beta(s)$. The often-quoted $\int_0^1 K(k)\,dk = 2G$ holds in the *modulus* convention only; in this paper's parameter convention—which is also the physical $d\rho$ measure of the co-area reduction below—$\int_0^1 K\,dm = 2$ and $\int_0^1 E\,dm = \frac{4}{3}$, both rational, so the convention-free reason G enters is the χ_{-4} theta series, not those integrals [**MEASURED;** `test_paper59_theta_chi4.py`]), neither of which lies in $\{1, \pi, K(\frac{1}{2}), P_8\}$. The conductor-4 arithmetic is thus carried by the fibre period at *every* fibre—a property of the family, not of the $\tau = i$ point—and the physical Bessel twist sharpens it: the decay rate $A = \sqrt{c_1} + \sqrt{c_2} = \sqrt{u}\,(1 + \sqrt{\rho})$ makes $\sqrt{\rho} = \sqrt{1-\lambda} = \theta_4^2/\theta_3^2$—a $\Gamma(4)$-level modular function—physical, promoting the level from $\Gamma(2)$ to $\Gamma(4)$, whose field is $\mathbb{Q}(\mu_4) = \mathbb{Q}(i)$, with $4 = |\mathrm{disc}\,\mathbb{Q}(i)| = \mathrm{cond}\,\chi_{-4}$ [**MEASURED;** `test_paper59_theta_chi4.py`]. In the corrected ring $\{\pi,\, K(\frac{1}{2}),\, 1/K(\frac{1}{2}),\, G\}$ the weight-≤ 2 search is decisively negative, and the period-only weight-≤ 3 search is negative, so the finite closed form, should it exist, is *weight-three and requires* G—and it is then the closed form of the $D \to 0$ *shadow*, the residual question being whether the physical $D = 1$ value itself collapses onto that ring element (since answered in the negative at clean heights by the 64-digit decoy-calibrated search above). This sharpened target also recasts the earlier merely-"consistent-with" negative as a ring-incompleteness artifact rather than, by itself, evidence of non-classicality: the non-classicality is carried separately, by the fibre's irregular twist—established symbolically at the operator level in Paper 59 (Newton polygon, Poincaré rank one) and witnessed numerically below—not by the integer-relation search.

[**OBSERVATION**] This places the three-center integral inside the elliptic (genus-one) layer of the cosmic-Galois / mixed-elliptic-motive structure that the Geometric Vacuum program's Tannakian substrate reaches for [10, 21]—the concrete elliptic period the mixed-Tate substrate could not itself supply. [**OPEN**] The natural next step is therefore to place it in the Bessel-moment period algebra—to test whether it obeys a Broadhurst–Roberts quadratic relation or reduces to a critical L-value of the associated modular form [8, 16]—of which the explicit $\Gamma(2)$ iterated-Eisenstein realization [13, 14] captures only the regular $D \to 0$ shadow, both of a piece with the finite closed form of Paper 59. We note the level is $\Gamma(2)$—the classical *universal* Legendre level, below the two-loop sunrise's $\Gamma_1(6)$; landing on the Legendre family is expected for a four-branch-point curve, so the content here is the rational modulus map and the in-domain complex-multiplication fibres, not an exotic level.

[**OBSERVATION**] The route to that closed form is now specific. The fibre period is the $\Gamma(2)$ modular quantity $K(m) = \frac{\pi}{2}\,\theta_3(0,q)^2$ at $\tau = iK'(m)/K(m)$ (with $\lambda(\tau) = m$, both validated to 40 digits), so the modulus integration pulls back to the τ-plane with the weight-two Jacobian in closed form, $d\lambda/d\tau = i\pi\,\theta_2^4\theta_4^4/\theta_3^4 = i\pi\,\lambda(1-\lambda)\theta_3^4$ (the classical modular-lambda derivative; leading q-expansion $i\pi(16q - 256q^2 + 2112q^3 - \cdots)$) [**MEASURED, $\sim 10^{-32}$ finite-difference + symbolic;** `test_paper59_bd_jacobian.py`]. The *physical* fibre, however, carries the $D = 1$ exponential (Bessel) twist the pure period lacks—it degenerates to $K_0(1)$, not a modular period, and this irregularity is measured directly at the level of the master period itself: its large-D asymptotic series is Gevrey-one (the optimal-truncation order grows linearly with D and the series then diverges) with Borel radius 2, matching the dominant singularity of the connection of Paper 59 [**MEASURED;** `test_paper59_resurgence_corner.py`, **with the irregular-connection witness** `test_L4_irregular_at_infinity`]—so the integrated observable is a $\Gamma(2)$ *resurgent Lambert series* rather than a bare modular period, the pure $\Gamma(2)$ multiple modular value being its regular $D \to 0$ shadow. This is exactly the class for which Broadhurst and Dorigoni give the iterated-Eisenstein / Lambert-series machinery for sunrise and banana integrals [3]. With the Jacobian above in closed form the length-two-to-length-one reduction is now explicit: a co-area change of variables (modulus $\rho = c_2/c_1$ outer, scale $u = c_1$ inner) collapses the observable to a single $X(2)$-locus integral $T2 = (16/\pi)\int_0^1 \Phi(\rho)\,d\rho$, folded by the exact symmetry $\Phi(\rho) = \Phi(1/\rho)/\rho^2$ [**MEASURED, reduction 2.2×10^{-9} vs the anchor, symmetry 5×10^{-29};** `test_paper59_coarea_reduction.py`]. The twist whose Eichler integral this is, is the *scale-integrated two-mass* fibre $\Phi(\rho)$—the one-mass $N(D) = \int_1^\infty e^{-Dx}/\sqrt{Q}\,dx$ is only its $\rho \to 0$ slice—and its cusp sits at the diagonal $c_1 = c_2$, where the curve degenerates to genus zero, so the leading Fourier–Whittaker (log) coefficient is the genus-zero diagonal value $A = J(\frac{1}{4}, \frac{1}{4}, 1)/4$ (with the exact relation $b_1 = -A$), one transcendence level below the observable. The cusp expansion $\Phi = \sum_m \left[b_m\lambda^m \ln\lambda + e_m\lambda^m\right]$ is *asymptotic*: its leading reliable terms overshoot (68% $\to$ 51% of $T2$, non-monotone), which excludes any convergent low-order closure and leaves the *Borel–Lambert resummation* to the second cusp—the Stokes / median-summation data one level beyond the coefficients (the BD $\sum_n a(n)q^n/(1-q^n)$ form)—as the single remaining specialist step.

[**SYMBOLIC + MEASURED**] The fold is itself a modular statement, and it locates the observable's modular home one level lower than the family's. With $c = s(1-s)$ invariant under $s \mapsto 1-s$, the full symmetry group of the Feynman square acts on the modulus $\rho = c_2/c_1$ through exactly $\{\mathrm{id},\ \rho \mapsto 1/\rho\}$; the physical $s \leftrightarrow t$ density exchange is $\rho \mapsto 1/\rho$, i.e. $\lambda \mapsto \lambda/(\lambda-1)$, the coset $T\Gamma(2)$, whose elliptic representative $M = \left[\begin{smallmatrix} 1 & -1 \\ 2 & -1 \end{smallmatrix}\right]$ satisfies $M^2 = -I$ and $M \equiv T \pmod 2$, and $\langle \Gamma(2), M\rangle = \Gamma_0(2)$. The folded co-area integral $\int_0^1 \Phi\,d\rho$ is therefore an integral over a fundamental domain for the extended group: *the observable lives on* $X_0(2) = X(2)/\langle\iota\rangle$, *not on* $X(2)$ (`tests/test_paper59_gamma0_2.py`). Two consequences, one positive and one deflationary. Positive: $\Gamma_0(2)$ has a unique elliptic point, of order two with $M^2 = -I$—disc -4, $j = 1728$, CM by $\mathbb{Z}[i]$—so the conductor-4 arithmetic attaches to the observable's modular home canonically, complementing the fibre-independent θ_3^2 leg above. Deflationary: that elliptic point sits at $\lambda = 2$ ($\rho = -1$), the ramification point of $X(2) \to X_0(2)$, *off* the physical contour $\rho > 0$; the involution's physical fixed locus is the diagonal $\rho = 1$, the pure-Tate cusp; and $\rho \mapsto 1-\rho$ is not induced by any physical symmetry, so the CM fibre $\rho = \frac{1}{2}$ is one member of the $s \leftrightarrow t$ orbit $\{\frac{1}{2}, 2\}$—a visited fibre, not a pinned one. (That a $\mathbb{Z}_2$ symmetry of a modular family pins a $\mathbb{Z}[i]$ point at its elliptic fixed locus is $\mathrm{PSL}_2(\mathbb{Z}) \cong \mathbb{Z}_2 * \mathbb{Z}_3$ torsion—order two forces disc -4, order three disc -3—so the arithmetic content of the involution is one bit, not a mechanism.)

[MEASURED] Two literature templates fix that step precisely and, between them, locate the one genuinely-open corner. A single-scale reduction of exactly this kind—a Bessel moment whose argument is a modular function, equated to a weight-two modular form and thence to an Eichler integral of a higher-weight form—is Zhou's [15] (e.g. $\int_0^\infty J_0(\alpha(z)t)\, I_0(t) K_0(t)^3\, t\, dt = \frac{\pi^2}{16} Z_{6,3}(z)$ at level $\Gamma_0(6)$, and the weight-four Eichler representation of $\int I_0 K_0^4$), whose special values close to complex-multiplication Γ-values or critical L-values. The resummation of the resulting cusp series across the two Fricke-dual cusps is Broadhurst–Dorigoni's, both in the divisor-sum form they give for sunrise/banana integrals [3] and—closer to our setup—in the character-twisted precursor [4], whose transseries relates the value at one cusp to the same Lambert family at the Fricke-dual argument $\tau \to -1/(r_1 r_2 \tau)$, median-resummed with algebraic (character) Stokes data; the general engine—a Borel tower whose Stokes constants are L-function coefficients, resummed to a quantum modular form—is the modular-resurgence paradigm of Fantini–Rella [5] and the Eichler-integral median resummation of McSpirit–Rolen [6]. Our object sits one axis beyond each: the fibre of Paper 59 is a *two-scale* ($a_1 \neq a_2$) Bessel moment where Zhou's are single-scale, and its cusp coefficients are Bessel–Fourier–Whittaker coefficients rather than the Dirichlet divisor sums those resummations twist by. The distinction is concrete already at the leading coefficient: the diagonal value $A = J(\frac{1}{4}, \frac{1}{4}, 1)/4 = 0.0790540321168\ldots$ is itself a single-scale Bessel moment—by the K_0 identity of Paper 59 and its t-derivative $\int_0^\infty \cos(kb)\, e^{-t\sqrt{ck^2+1}}\, dk = 2t\, K_1(\sqrt{t^2+4b^2})/\sqrt{t^2+4b^2}$, A reduces (exactly) to a σ-integral of K_1 at argument $\sqrt{t^2+4\sigma^2}$—and a guarded, decoy-controlled integer-relation search finds *no* low-height closure for it in the weight-one two-center ring $\{\pi, e^{-a}, K_0, K_1, E_1\}$ at the natural arguments $a \in \{2, 2\sqrt{2}\}$ **[MEASURED;** `test_routeC_momentum.py::test_paper59_diagonal_A`]. So even the leading cusp datum is an irreducible Bessel moment, not an elementary or divisor-sum quantity—consistent with a resurgent Lambert series whose individual coefficients need not be elementary for the resummed value to be a clean $\Gamma(2)$ period. **[OPEN]** The exact remaining object is therefore the weight-two Eisenstein Eichler integral over $X(2)$ against this two-scale Bessel twist—the two-scale generalization of the single-scale Bessel-moment theory of Fresán–Sabbah–Yu [8], itself open—covered by none of these templates individually but pinned by all of them jointly.

[MEASURED] The second-cusp resurgence itself is now characterized in closed form. The scale-integrated twist's one-mass expansion ($c_2 \to 0$, one scale collapsing) is Gevrey $(2n)!$-type, and its dominant Borel singularity—the Stokes-data *location*—is $z^* = -(\sqrt{c_1} \mp ib)^2 = (b^2 - c_1) \pm 2i\sqrt{c_1}\, b$, with $|z^*| = c_1 + b^2$ and trans-series action $\sqrt{z^*} = b + i\sqrt{c_1}$ (the source separation b and the scale $\sqrt{c_1}$ fused into one complex action). It is a $(z^* - w)^{3/2}$ branch (exponent $p = -\frac{5}{2}$, measured cleanly) with amplitude in *closed form*, $2|A| = (c_1 + b^2)^{3/2}/(4\sqrt{\pi}\,\sqrt{c_1}\, b)$ (algebraic $\times 1/\sqrt{\pi}$, the $1/\sqrt{\pi}$ from the $(1+\varepsilon)^{-5/2}$ boundary branch)—derived from the moment tail rate $\sqrt{c_1} - ib$ and validated at three parameter points (fit-to-closed-form ratio 0.988–0.989, a uniform finite-order deficit converging to 1). This is the two-scale analog of the algebraic-Stokes closure of $N(D)$ in Paper 59, and the median-Borel resummation across the (complex, off-contour) singularity pair is unambiguous. **[OPEN]** The residual step to the finite value is the integration of this algebraic-Stokes trans-series over the Feynman family (the inner scale) and the modulus—the concrete form the two-scale Broadhurst–Dorigoni resummation takes here. Brute high-precision quadrature is not a route *in the* (s,t) *or co-area frames* (complex-singularity- and fibre-cost-limited there); the (k,w) refactorization Eq. (2) removes exactly that obstruction and supplies the *digits*, while the finite *closed form* remains this resummation question.

[MEASURED] *The Stokes location is a complexified light cone.* In position space each packet is analytic in b^2 with branch points at the endpoint of its source-time superposition, where the Euclidean interval closes, $b^2 + p_i^2 = 0$; the j_0 smearing moves the singularity to the endpoint pinch $\beta^* = \pm|\boldsymbol{W}| \pm ip_1$, so the Borel location above is the squared complexified interval, $z^* = (\beta^*)^2$, $|z^*| = p_1^2 + |\boldsymbol{W}|^2$, of which $-(\sqrt{c_1} \mp ib)^2$ is the $D = 1$ specialisation $p_1 = \sqrt{c_1}$; the trans-series action $a_2\sqrt{z^*}$ is the heavy mass times that interval. Moment growth of the position-space fibre confirms modulus and phase at eight $(c, |\boldsymbol{W}|, D)$ points, including $D \neq 1$, where the interval form and the D-independent form differ by 35% **[MEASURED,** 0.6–5% **from twelve coefficients]**. Two consequences. The median resummation is unambiguous because a Euclidean interval never vanishes at real separation: the absence of a Stokes ambiguity is Euclidean positivity. And the $(s,t) \to (0,0)$ corner is the coincidence (ultraviolet) point, where the interval closes and the Borel singularities collide at the origin—which is why that corner, and only that corner, obstructs the family integration. There the masses diverge at fixed ratio, so the modulus survives and the Duffy angle *is* the modulus; the leading behaviour is the ultraviolet power count $J = \sigma^3 A(a) + O(\sigma^5)$, $\sigma^3 = m_1^{-2} m_2^{-2} m$, so the $\rho^{3/2}$ non-analyticity is an odd mass dimension and carries no π. At the three corners with $|\boldsymbol{W}| \neq 0$ the oscillation survives, a Dirichlet limit supplies one further power and an explicit π, and the integer-leading behaviour is now closed: $J = \sigma^4 \frac{\pi}{2} G(1)^2 \alpha(1-\alpha)/|\boldsymbol{W}| + O(\sigma^5)$ with $G(1) = 7/e$ **[MEASURED,** 10^{-4} **with** $O(\sigma^2)$ **approach, four corner/angle combinations;** `test_paper59_euclidean_dictionary.py`].

[OBSERVATION] Read against the two-layer taxonomy, the object separates on two kinematic axes: the mass split turns on the elliptic (genus-one) level and the Euclidean-time displacement turns on the irregular level—the curve carries no D, and the large-D series is Gevrey-one at every ρ, including the genus-zero slice $\rho = 0$ where the fibre is $K_0(D)$—while the modulus then also grades the irregular data through the Borel radius $\min(2,\, a_2/a_1)$, the smallest difference of the four exponential rates. Every π in the object comes from the momentum-space measure or from integration over the momentum half-line; the non-compact Euclidean-time direction contributes no π but the exponential and its Stokes data, consistent with the compactification reading of Paper 35 [20]. (In Euclidean signature the labels are structural, not metric: D is the shift each propagator carries, b the shared coordinate that is integrated out; the temporal reading is offered as a reading.)

[OBSERVATION] The algebraic-Stokes closures of this paper sit inside a pattern that now spans the corpus's two-electron integral classes. Four objects have had their resurgent data computed: the Paper-18 exchange-class seed $e^a E_1(a)$ (Borel transform $1/(1+\zeta)$, one simple pole, Stokes constant $2\pi i \cdot 1$); the Paper-58 hybrid two-center class $\{E_1, \ln\}$ (all Borel singularities at $\pm a_d$, the far-center exponent; Stokes constants $2\pi i \times \mathbb{Q}(\sqrt{d})$, extracted blind from large-order coefficients to 35–81 digits and matched by exact integer relations; and the sector constants cancel pairwise, so the physical integral is *cut-free* in R—the median resummation *is* the closed form, `tests/test_paper59_resurgent_skeleton.py`); the one-mass fibre $N(D)$ of Paper 59 (algebraic over $\mathbb{Q}(\rho)$); and the second-cusp triple above (algebraic $\times 1/\sqrt{\pi}$). In all four the connection data is algebraic over the parameter field up to a π-power that is itself fixed by the Borel singularity type (simple pole $\to 2\pi i$; square-root branch $\to \pi^0$; three-halves branch $\to 1/\sqrt{\pi}$)—nothing in the Stokes data is free. What varies is the *boundary period*: absent for the seed; $\ln \Lambda$ with $\Lambda \in \mathbb{Q}$ the multiplicative cross-ratio of the four Borel actions—i.e. *forced by the skeleton*—for the hybrid class; the elliptic $K(1-\rho)$, *not* determined by the Stokes data, for $N(D)$; and the open $\Gamma(2)$ object for $T2$. The forced$\to$free transition of the boundary period thus sits between weight one and genus one—the same seam at which the closed-form program of this paper opened. The named next test—the exchange class, whose R-dependent logarithm makes its Borel singularity logarithmic rather than polar—has since been run and the pattern stands at **five for five**: the exchange kernel collapses to a single-sector reduced normal form whose Borel transform is the closed-form, charge-neutral sum $\mathfrak{B}(\xi) = \frac{1}{2ab} \sum_j q_j (\xi - \xi_j) \ln(\xi - \xi_j)$ over the four rational positions, with *integer* Stokes charges $q_j \in \{-1, +1, +1, -1\}$ (no radical needed, unlike the hybrid's $\sqrt{d}$); the boundary log is forced even more tightly than the hybrid's cross-ratio—its argument is $\kappa = \prod_j \lambda_j^{q_j} = 2ab/(a+b)$, the exponents *being* the Stokes charges; and Euler's γ never exists in the Borel plane at all: the R-dependent log moves one Borel singularity onto the sector's own origin (pole $\to u \ln u$ branch), and γ is exactly the bookkeeping cost of writing that origin singularity in the R variable—which finally tags the γ of the weight-one seed set as coordinate bookkeeping, not a projection transcendental. The object is Borel-summable with no lateral ambiguity yet genuinely multivalued in complex R (the price of the R-dependent log, costing no new transcendental), where the hybrid class was single-valued (`tests/test_paper59_resurgent_skeleton.py`). The π-power law refines accordingly: an *integral* local Borel exponent (pole *or* log branch) fixes $2\pi i$; a half-integral one fixes π^0 or $1/\sqrt{\pi}$. (Read against the two-layer taxonomy this is a candidate refinement—a resurgent-skeleton grading *within* the transcendental layer—recorded as an observation, now on five objects; the residual open leg is the heteronuclear infinite τ-sum.)

III. THE BESSEL-MOMENT PERIOD ALGEBRA: MASTER FAMILY AND QUADRATIC RELATIONS

Section II places the integrated observable in the Bessel-moment class; we now make that placement quantitative by exhibiting the period algebra it inhabits. The rank-four irregular connection of Paper 59 has four master periods, the Laplace transforms

$$s_c(D) = \int_{\gamma_c} \frac{e^{-Dx}\, dx}{\sqrt{Q(x)}}, \qquad Q(x) = (x^2-1)(\rho x^2 + 1 - \rho), \tag{3}$$

over the four independent cycles γ_c joining the branch points $\{\pm 1, \pm i\sqrt{(1-\rho)/\rho}\}$: the physical moment s_K (the cut $[1,\infty)$, which is $N(D)$ itself, $\sim e^{-D}$), its companion s_I ($[-1,1]$, $\sim e^{+D}$), the imaginary-axis period s_J ($\sim \cos\omega D$), and the sector s_Y carrying the $D \ln D$ non-analyticity. **[MEASURED]** The three regular masters satisfy the Picard–Fuchs equation of Paper 59 to $\sim 10^{-15}$ across ρ and D (each annihilated: $s_K \sim 10^{-18}$, $s_I \sim 10^{-17}$, $s_J \sim 10^{-15}$).

The Wronskian. **[SYMBOLIC]** The operator of Paper 59 is formally self-adjoint, so its differential Galois group lies in Sp_4, and Abel's identity—with the subleading coefficient ratio $p_3/p_4 = 2/D$—fixes the Wronskian of the master matrix exactly as a pure monomial,

$$W(D) = W_0\, D^{-2}: \tag{4}$$

the D-dependence of the determinant is the pure monomial D^{-2}, the elliptic analogue of the classical $W[K_0, I_0](D) \propto 1/D$ and the two-scale counterpart of the Wrońskian factorizations behind the Broadhurst–Mellit determinant formulae [17]. *Correction, 2026-09-07* `[retracted 2026-09-07: p61-w0-transcendence-cancels]`. Earlier versions added that "the transcendence of the individual masters cancels in their determinant". That is *false*, and the value below is what establishes it: Abel's identity fixes only the D-dependence, and the constant W_0 is *never computed* in this paper or in its backing test, which normalises it to 1 by construction. It is now computed, from the same branch-point data that fixes B below. With $Q(x) = (x^2-1)(\rho x^2 + 1 - \rho)$ and branch points $\{\pm 1, \pm i\omega\}$, $\omega^2 = (1-\rho)/\rho$: $\sum_c x_c = 0$ cancels the exponentials; $\prod_c D^{-1/2} = D^{-2}$ reproduces Abel's exponent by an independent route; $\prod_c Q'(x_c) = -16\omega^2$, so $\prod_c A_c = \pi^2/(4i\omega)$; and the Vandermonde $\prod_{a<b}(x_b - x_a) = 4i\omega/\rho^2$ cancels the ω, leaving

$$W_0 \;=\; \frac{\pi^2}{\rho^2} \qquad \textbf{[SYMBOLIC]}, \tag{5}$$

up to the thimble ordering/orientation sign this paper already riders for B (the sign is the branch of $\sqrt{1-\rho}/\sqrt{\rho-1} = -i$ on $0 < \rho < 1$). So π^2 does *not* cancel — it is squared — and W_0 is ρ-dependent, which is what makes the withdrawn sentence false rather than merely unsupported. The withdrawal is a net gain: an exact value replaced a false cancellation.

Quadratic period relations. **[MEASURED, 25 digits]** Self-adjointness makes the Lagrange bilinear concomitant $B[s_a, s_b]$ constant in D; evaluated on the masters it is moreover independent of the modulus ρ and lands on rational multiples of π,

$$B[s_K, s_I] = -\pi, \qquad B[s_K, s_J] = 0, \qquad B[s_I, s_J] = 2\pi, \tag{6}$$

verified to 25 digits at $D = 1$ across the tested ρ, and D-independent by the self-adjoint Lagrange identity. These are the Broadhurst–Roberts / Fresán–Sabbah–Yu quadratic relations between Bessel moments [8, 16], here for the $\Gamma(2)$ (Legendre) family—the elliptic lift of $W[K_0, I_0] = 1/D$. Structurally the concomitant is the de Rham–Betti intersection pairing of the thimble cycles times the branch-point half-residue π (the square-root (-1)-monodromy of $1/\sqrt{Q}$ at each simple branch point, the elliptic lift of the $1/\pi$ normalizing I_0 and of $W[K_0, I_0] = 1/D$), so $B = \pi\, J$ with J an integer intersection form.

[SYMBOLIC] *The intersection form, identified.* We close the identification of J from the branch-point asymptotics—no numerics enter in this leg. Near a simple branch point $Q \sim Q'(x_c)(x - x_c)$, so each master has the leading Laplace behaviour $s_c(D) \sim e^{-x_c D}\sqrt{\pi/(Q'(x_c)\, D)}$ (the half-order integral $\int_0 e^{-Dt} t^{-1/2}\, dt = \Gamma(\tfrac12)/\sqrt{D}$); since the concomitant is exactly constant in D, its value equals its $D \to \infty$ limit, which these leading terms capture (the subleading $O(1/D)$ corrections vanish in the limit). In the Lefschetz-thimble basis $\{\gamma_{+1}, \gamma_{-1}, \gamma_{+i\omega}, \gamma_{-i\omega}\}$ —one thimble per branch point, the basis in which the monodromy M_0 of Paper 59 is the exact integer matrix—two facts follow. *Block-diagonality:* $s_a s_b \sim e^{-(x_a+x_b)D}$, so a nonzero *constant* forces $x_a + x_b = 0$; the branch points pair as $\{+1, -1\}$ (the real K_0/I_0 sector) and $\{+i\omega, -i\omega\}$ (the imaginary J_0/Y_0 sector), and every cross-sector pairing ($x_a + x_b \neq 0$) vanishes identically. *Plane-equality and the unit* π*:* on a within-sector pair the leading concomitant is $B[s_x, s_{-x}] = 2x\,(-2\rho x^2 + 2\rho - 1)\, A_x A_{-x}$ with $A_x A_{-x} = \pi/\sqrt{Q'(x)Q'(-x)}$, and the exact identity $Q'(x)Q'(-x) = -4x^2(2\rho x^2 - 2\rho + 1)^2$ cancels the algebraic factor completely, leaving

$$B[s_x, s_{-x}]^2 = -\pi^2 \qquad \text{independently of } x \text{ and } \rho, \tag{7}$$

so *both* symplectic planes carry the same pairing, of magnitude exactly π (the $\pi = \Gamma(\tfrac12)^2$ of the two branch-point half-integrals). Hence

$$J = \Omega := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad B = \pi\,\Omega, \tag{8}$$

the canonical block symplectic form; in the period-cut basis this is the real $-\pi, 0, 2\pi$ measured above (a numerical thimble normalization instead reads $B = \pi(\rho i)\Omega$, the same Ω). This Ω is *forced*, not fitted: the antisymmetric forms preserved by the monodromy, $M_0^{\mathsf T} J M_0 = J$, constitute a four-parameter family, and imposing the concomitant's vanishing cross-sector pairings selects *exactly* $\mathbb{Z}\,\Omega$. It is unimodular ($\det\Omega = 1$, hence nondegenerate), and $M_0^{\mathsf T}\Omega\, M_0 = \Omega$ holds identically over $\mathbb{Z}$, so the *monodromy* group of the Picard–Fuchs equation of Paper 59 lies in the *integral* symplectic group $\mathrm{Sp}(\Omega, \mathbb{Z}) = \mathrm{Sp}_4(\mathbb{Z})$—the concrete, integral sharpening of the self-adjointness that placed the Galois group in $\mathrm{Sp}_4(\mathbb{C})$. *Correction, 2026-09-07.* Earlier versions of this paper said the *differential Galois group* lies in $\mathrm{Sp}_4(\mathbb{Z})$. That is false: $\mathrm{Sp}_4(\mathbb{Z})$ is discrete, so a Zariski-closed subgroup of $\mathrm{GL}_4(\mathbb{C})$ contained in it would be finite, forcing every solution to be algebraic—contradicting the irregularity of the connection and the exponential torus $(\mathbb{C}^*)^2$ that Paper 59 establishes inside the local Galois group. The integrality is a statement about the Betti lattice, i.e. the monodromy representation, which is what M_0 is; Paper 59 states the Galois containment correctly as $\mathrm{Sp}_4(\mathbb{C})$. Nothing else changes: $B = \pi\,\Omega$, the forcing of Ω, and the integral structure all stand. *Tier scope.* The asymptotic identification above is numerics-free, but every clause invoking M_0—the forcing argument and the $\mathrm{Sp}_4(\mathbb{Z})$ containment—inherits that matrix's **[SYMBOLIC + MEASURED]** status: M_0 is an exact integer certificate rather than a floating-point fit, but it was obtained by computation, not derived in closed form. The three physical period cuts $\{[1,\infty), [-1,1], i[-\omega,\omega]\}$ realize only a rank-two sub-block of $\pi\Omega$ (the measured $-\pi, 0, 2\pi$); it is the fourth, Y_0-sector master—the one carrying the $D \ln D$—that completes the pairing to the full nondegenerate rank-four form. The symbolic identification of the intersection form is thus complete: $B = \pi\,\Omega$, the Broadhurst–Roberts / Fresán–Sabbah–Yu quadratic relations realized for the $\Gamma(2)$ family, backed by `tests/test_routeC_momentum.py` (`test_intersection_form_pi_and_planes_are_symbolic` for Eq. (7) and the block structure, `test_intersection_form_is_forced_canonical_symplectic` for the M_0-forcing, `test_intersection_form_period_cut_values` for the $-\pi, 0, 2\pi$ values, and `test_paper59_bessel_moment_algebra.py::test_paper59_period_pair_10_minus_pi_0_2pi` witnessing them to the quoted 25 digits).

Not a cusp-form L-value. **[OBSERVATION]** Because $M_\bullet(\Gamma(2)) = \mathbb{C}[\theta_2^4, \theta_4^4]$ is free on two weight-two generators, $\dim S_k(\Gamma(2))$ vanishes for $k = 2, 4$ and first reaches 1 at $k = 6$ (the cusp form $\theta_2^4\theta_3^4\theta_4^4$). The observable's transcendental weight is at most three (length ≤ 2 over $X(2)$, from the two Feynman integrations), whereas a weight-six cusp form's critical L-values sit at motivic weight five; and the period pairing just exhibited is Eisenstein-flavoured (pure π-rationals), not cuspidal. The integrated three-center integral is therefore *not* a critical L-value of a $\Gamma(2)$ cusp form: it is an Eisenstein / complex-multiplication period, built from π and the CM-fibre Γ-values of Sec. II ($K(\frac{1}{2}) = \Gamma(\frac{1}{4})^2/4\sqrt{\pi}$, the discriminant-8 period, and their kin). **[MEASURED]** A guarded integer-relation search (with a same-magnitude decoy control) confirms the value is not a low-height combination of the small weight-≤ 2 disc-4 Eisenstein/CM ring, nor of the small weight-≤ 1 disc-8 ring $\{1, \pi, \varpi, P_8\}$ (P_8 the discriminant-8 period)—both *decisive* at the conservatively cross-validated ~ 16 digits: two structurally independent evaluators (the spectral corner of Sec. II and a decay-scaled-fibre quadrature) agree to 16 digits, the corner evaluator extending single-method to ~ 19; on the decisive rings the relation height is far above low and tracks the decoy at every tested precision. The wider searches—through weight three in disc-4, and the disc-8 ring at weight two—remained only *consistent* with the negative: those dimension-ten bases are over-determined at this precision (target and decoy both turn up small but precision-*unstable* relations, changing from one precision to the next). The decision has since been made at 64 digits via the (k, w) refactorization (Eq. (2)): *decisively negative* through weight three, including the Catalan-completed corrected ring, at height ≤ 10—so the finite closed form, if it exists at all, either has height > 10 in that ring or lives outside it.

This narrows the open problem sharply: the finite closed form, should it exist, is a reduction of the observable to π, the CM-fibre Γ-values, their Legendre *quasiperiods*, and the weight-two Eisenstein L-value $G = \beta(2)$ (Sec. II), within the Bessel-moment period algebra—a two-scale, shifted member of the Broadhurst–BBBG class [1, 2] evaluated over $X(2)$—and is neither an elliptic dilogarithm nor a cusp-form L-value. **[OBSERVATION]** Because the corrected-ring searches are now decisively negative through weight three at clean heights (64 digits, height ≤ 10), any such finite combination either belongs to the $D \to 0$ *shadow*—where the shadow reading of Sec. II places the weight-three G-requiring element—or sits at height > 10 in that ring; the physical value itself carries no measured conductor-4 content at reachable heights.

[1] D. H. Bailey, J. M. Borwein, D. Broadhurst, and M. L. Glasser, "Elliptic integral evaluations of Bessel moments and applications," *J. Phys. A* **41**, 205203 (2008); arXiv:0801.0891.

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